

```
%KAFKA_HOME%\bin\windows\zookeeper-server-start.bat  
%KAFKA_HOME%\config\zookeeper.properties
```

```
%KAFKA_HOME%\bin\windows\kafka-server-start.bat  
%KAFKA_HOME%\config\server.properties  
%KAFKA_HOME%\bin\windows\kafka-topics.bat --create -  
-bootstrap-server localhost:9092 --replication-factor 1 --  
partitions 1 --topic nov22
```

#ur not using next 2 lines in ur office anywhere its for only testing purpose

```
%KAFKA_HOME%\bin\windows\kafka-console-  
producer.bat --bootstrap-server localhost:9092 --topic  
nov22
```

```
%KAFKA_HOME%\bin\windows\kafka-console-  
consumer.bat --topic nov22 --from-beginning --  
bootstrap-server localhost:9092
```

```
%KAFKA_HOME%\bin\windows\kafka-topics.bat --list --  
bootstrap-server localhost:9092
```

here what is broker ,producer and consumer



Here's a **simple and clear explanation** of **Kafka Broker, Producer, and Consumer**, based on the commands you shared.

1. What is a Kafka Broker?

A **Kafka Broker** is a **Kafka server**.

- It stores messages inside **topics**.
- It receives data from **producers**.
- It sends data to **consumers**.

- You can have **1 broker** or **multiple brokers** (cluster).

In your command:

```
pgsql
```

[Copy code](#)

```
kafka-server-start.bat %KAFKA_HOME%\config\server.properties
```

👉 This starts **one Kafka broker** running on port **9092**.

✓ 2. What is a Kafka Producer?

A **Producer** is a client application that sends (produces) messages to Kafka.

Example:

```
css
```

[Copy code](#)

```
kafka-console-producer.bat --bootstrap-server localhost:9092 --topic nov22
```

👉 This command creates a **producer**.

- Whatever you type in the console gets **sent to the Kafka topic** nov22 .

In real projects:

- Producers are usually written in **Java, Python, Node, Go, etc.**
- Example: Microservice or app sending logs, events, transactions.

✓ 3. What is a Kafka Consumer?

A **Consumer** is a client application that reads (consumes) messages from Kafka.

Example:

```
css
```

[Copy code](#)

```
kafka-console-consumer.bat --topic nov22 --from-beginning --bootstrap-server loca
```

👉 This creates a **consumer**.

- It reads messages from the topic **nov22**.
- `--from-beginning` means it reads **all previous messages**.

In real projects:

- Consumers might be services that read data from Kafka and process it:
 - Fraud detection
 - Notifications system
 - ETL pipelines
 - Analytics

Putting it together with a simple diagram

css

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[Producer] → send messages → [Kafka Broker] → store/distribute → [Consumer]



Or with your example:

javascript

 Copy code

kafka-console-producer → (message) → Kafka Broker (9092) → kafka-console-consumer



Summary Table

Component	What it is	Your command
Broker	Kafka server storing data	kafka-server-start.bat
Producer	Sends data to Kafka	kafka-console-producer.bat
Consumer	Reads data from Kafka	kafka-console-consumer.bat

If you want, I can also explain:

- What is **Zookeeper**?
- How to run **Kafka without Zookeeper** (latest versions use KRaft)?
- How to install **confluent-kafka** Python library?
- Full **Kafka producer/consumer example in Python**.

Just tell me!



yes where is zookeeper here

Great question — let me show you **exactly where Zookeeper comes in** and why you *sometimes* need it.

✓ Where is ZooKeeper in your commands?

You are starting ZooKeeper with this command:

```
perl
```

Copy code

```
%KAFKA_HOME%\bin\windows\zookeeper-server-start.bat %KAFKA_HOME%\config\zookeeper
```



👉 This is ZooKeeper.

ZooKeeper must start **before Kafka broker** in older Kafka versions (≤ 2.8).

🧠 What does ZooKeeper do? (Simple Explanation)

ZooKeeper is a **coordination service**.

Kafka uses it to store:

- Broker information
- Controller election
- Topic metadata

- Partition leadership
- Consumer group offsets (old versions)

Think of ZooKeeper as:

```
rust
```

[Copy code](#)

```
Kafka's brain for coordination.
```

Kafka brokers cannot work alone in older versions — they need ZooKeeper.

✓ Your Flow (Using ZooKeeper Mode)

Here's the full flow based on your commands:

```
sql
```

[Copy code](#)

1 Start ZooKeeper

```
zookeeper-server-start.bat zookeeper.properties
```

2 Start Kafka Broker (depends on ZooKeeper)

```
kafka-server-start.bat server.properties
```

3 Create Topic

```
kafka-topics.bat --create ...
```

4 Produce messages

```
kafka-console-producer.bat ...
```

5 Consume messages

```
kafka-console-consumer.bat ...
```

So, ZooKeeper is step #1.

BUT... Latest Kafka does NOT use ZooKeeper

As of **Kafka 3.x and above**, Kafka supports **KRaft mode**, which replaces ZooKeeper.

That means:

- No ZooKeeper needed
- You start only the Kafka server
- Metadata is handled internally

If you're using **Confluent Platform** or **Kafka 3.6+**, you'll see a file named:

```
pgsql
```

[Copy code](#)

```
kafka-storage.sh
```

or

```
pgsql
```

[Copy code](#)

```
kraft-server-start.sh
```

Then your lifecycle becomes:

```
pgsql
```

[Copy code](#)

- 1 Format storage directory (once)
- 2 Start Kafka directly (no ZooKeeper)

? Want to check which mode YOU are running?

Tell me which Kafka version you installed.

Run:

```
css
```

[Copy code](#)

```
%KAFKA_HOME%\bin\windows\kafka-topics.bat --version
```

Copy-paste the output here, and I'll tell you:

- whether you need ZooKeeper
- or whether your Kafka can run in **KRaft mode** (no ZooKeeper needed)

If you'd like, I can also:

- Show how to run Kafka **without ZooKeeper**
- Give you a **correct Windows startup script**
- Explain **Confluent Kafka vs Apache Kafka**
- Show a simple **Python producer/consumer using confluent_kafka**

Just let me know!

